LAVKUSH KATIYAR

BACKEND DEVELOPER | PYTHON, FASTAPI & AI INTEGRATION

+91 7905837232 | worklavkush@gmail.com | Bengaluru
LinkedIn | Github | LeetCode | GeeksForGeeks| Portfolio

EXPERIENCE

STEP Intern at ThoughtWorks India **July 2025 – March 2026**

- Developed CLI-based software utilities using JavaScript and Deno, applying modular design, recursion, and structured input processing.
- Worked with HTTP client–server interactions using curl and fetch, examining request methods, responses, and communication behavior.
- Used Unix command-line tools and shell scripting to inspect, filter, and process data through repeatable workflows.
- Applied Deno.js fundamentals and modular backend design principles that I later extended to REST API development with Node.js and Express.js.

Trainee at Techpile Technology PVT LTD, Lucknow **July 2024 – September 2024**

- Gained hands-on experience in Python, Django, and web development fundamentals through guidence training and projects.
- Worked on backend logic, database integration, and basic application workflows under mentorship. Strengthened problem-solving skills and understanding of software development best practice

PROJECTS

Kairox – AI Workflow Automation Backend

(Python, FastAPI, Pydantic, SQLAlchemy, PostgreSQL, Alembic, JWT, Redis, Celery, Ollama, Docker) (May 2026 – Present)

- Developed versioned REST APIs for 6 areas: authentication, input processing, task management, dependencies, reminders, background jobs.
- Implemented JWT authentication, rate limiting, and circular-dependency prevention for secure, reliable task management.
- Designed SQLAlchemy and PostgreSQL data models, managing schema changes with Alembic migrations.
- Integrated Redis and Celery for asynchronous processing and Ollama for local LLM inference Openrouter for deployment LLM.

SupportPilot — AI Customer Support Agent

(Python, FastAPI, Pydantic, Sentence Transformers, PostgreSQL, pgvector, Pytest, Bruno, Ruff, GitHub Actions) (Aug 26 – Oct 26)

- Achieved 89% intent-classification accuracy using a defined taxonomy, structured Pydantic outputs, and iterative error analysis.
- Processed Kaggle's ~3-million-tweet Customer Support on Twitter dataset, reconstructing Apple Support conversations for retrieval.
- Implemented semantic retrieval using Sentence Transformers and pgvector to ground AI-generated responses in relevant historical cases.
- Built a repeatable evaluation workflow with labeled data, classification metrics, GitHub Actions CI, API validation, and rate limiting.

PromptCost — AI API Usage & Cost Analyzer

Python, FastAPI, Pydantic, PostgreSQL, SQLAlchemy, Alembic, Redis, pytest, Docker,Git, GitHub Actions (Sep 26 – Oct 26)

- Designed backend architecture tracking 4 AI API metrics: token usage, latency, failures, and estimated model costs.
- Defined usage-event models across 5 dimensions: provider, model, project, user, and time period.
- Designed cost-analysis logic using model-specific input and output token pricing for accurate usage estimates.
- Planned 3 scalability mechanisms: usage aggregation, caching, and asynchronous processing for analytics and monitoring.

Financial Record Management System

Node.js, Express.js, PostgreSQL, Prisma ORM, JWT, Zod | April 2026 – May 2026

- Implemented 2 security layers: JWT authentication and role-based access control (RBAC).
- Modeled relational financial data in PostgreSQL using Prisma ORM, linking records with categories.
- Applied 3 backend practices: layered architecture, Zod validation, and centralized error handling to enforce API contracts.

EDUCATION

Bachelor of Computer Applications (BCA)
University of Mysore **Pursuing (Online)** **2025 - 2028**

Diploma in Information Technology
Government Polytechnic Lucknow **2022 - 2025**

ACHIEVEMENTS

Coding Practice: Solved 250+ problems across LeetCode (200+) and GeeksforGeeks (50+).
Advent of Code: Completed programming challenges in 2019 and 2025.

Technical Skills

Languages: Python, JavaScript (Node.js, Deno), SQL
Backend: FastAPI, Express.js, REST APIs, Pydantic, SQLAlchemy, Prisma ORM
Databases: PostgreSQL, MySQL, pgvector, Alembic
AI & LLMs: Ollama, OpenRouter, LLM Integration, Prompt Engineering, Structured Outputs, Embeddings, Semantic Search, Model Evaluation
Frontend: React, Redux Toolkit, HTML5, CSS3, Tailwind CSS
Tools & Infrastructure: Git, GitHub, GitHub Actions, Docker, Docker Compose, Linux, Redis, Celery, Render, Vercel, Supabase, Postman Bruno, pytest, Ruff